

Dionex_COMPASS_TEMPEST

Stephanie J. Wilson

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Daily Set up

Read in and Format the raw data - change wd & file names (Make sure files are UTF-8 encoded)

```
# SULFATE DATA:
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Sdat <- read.table("Raw Data/COMPASS_TEMPEST_20230608_S04.txt", sep='\t' , header=T, skip=3)  
head(Sdat)
```

```
##      X          X.1                X.2 IC.S04 IC.S04.1 IC.S04.2 IC.S04.3 IC.S04.4  
## 1 1 Lab Blank          Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 2 2 Lab Blank          Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 3 3 Lab Blank          Unknown    n.a.    n.a.    n.a.    n.a.    n.a.  
## 4 4 Standard 1 Calibration Standard 4.343 0.4565 6.17 0.5324 2.53  
## 5 5 Standard 2 Calibration Standard 4.343 0.9668 6.58 1.1275 5.32  
## 6 6 Standard 3 Calibration Standard 4.33 1.9262 7.25 2.2463 10.35
```

```
## Only keep the columns that we need
```

```
Sdat <- Sdat[ ,c(2,5,7)] # dont need this here  
head(Sdat)
```

```
##          X.1 IC.S04.1 IC.S04.3  
## 1 Lab Blank    n.a.    n.a.  
## 2 Lab Blank    n.a.    n.a.  
## 3 Lab Blank    n.a.    n.a.  
## 4 Standard 1 0.4565 0.5324  
## 5 Standard 2 0.9668 1.1275  
## 6 Standard 3 1.9262 2.2463
```

```
## Name the columns correctly
```

```
colnames(Sdat) <- c("Sample_ID", "S04_ppm", "S04_Area")  
Sdat$Sample_ID <- as.factor(Sdat$Sample_ID)  
Sdat$S04_ppm <- as.numeric(Sdat$S04_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat$S04_Area <- as.numeric(Sdat$S04_Area)
```

```
## Warning: NAs introduced by coercion
```

```
Sdat <- as.data.frame(Sdat)
head(Sdat)
```

```
##      Sample_ID S04_ppm S04_Area
## 1 Lab Blank      NA      NA
## 2 Lab Blank      NA      NA
## 3 Lab Blank      NA      NA
## 4 Standard 1  0.4565  0.5324
## 5 Standard 2  0.9668  1.1275
## 6 Standard 3  1.9262  2.2463
```

```
#Chloride data
```

```
## Read in raw data file from Dionex - copied and saved as a txt
```

```
Cldat <- read.table("Raw Data/COMPASS_TEMPEST_20230608_Cl.txt", sep='\t' , header=T, skip=3)
head(Cldat)
```

```
##      X      X.1      X.2 IC.Cl IC.Cl.1 IC.Cl.2 IC.Cl.3 IC.Cl.4
## 1 1 Lab Blank      Unknown  n.a.    n.a.    n.a.    n.a.    n.a.
## 2 2 Lab Blank      Unknown  n.a.    n.a.    n.a.    n.a.    n.a.
## 3 3 Lab Blank      Unknown  n.a.    n.a.    n.a.    n.a.    n.a.
## 4 4 Standard 1 Calibration Standard 3.417  5.7300  93.70  8.0833  45.41
## 5 5 Standard 2 Calibration Standard 3.417  11.3295  93.26  15.9825  88.56
## 6 6 Standard 3 Calibration Standard 3.413  20.3441  92.64  28.6992  159.19
##      IC.Cl.5
## 1      n.a.
## 2      n.a.
## 3      n.a.
## 4      BM
## 5      M
## 6      BM
```

```
## Only keep the columns that we need
```

```
Cldat <- Cldat[ ,c(2,5,7)]
head(Cldat)
```

```
##      X.1 IC.Cl.1 IC.Cl.3
## 1 Lab Blank  n.a.    n.a.
## 2 Lab Blank  n.a.    n.a.
## 3 Lab Blank  n.a.    n.a.
## 4 Standard 1  5.7300  8.0833
## 5 Standard 2  11.3295  15.9825
## 6 Standard 3  20.3441  28.6992
```

```
## Name the columns correctly
```

```
colnames(Cldat) <- c("Sample_ID", "Cl_ppm", "Cl_Area")
Cldat$Sample_ID <- as.factor(Cldat$Sample_ID)
Cldat$Cl_ppm <- as.numeric(Cldat$Cl_ppm)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat$Cl_Area <- as.numeric(Cldat$Cl_Area)
```

```
## Warning: NAs introduced by coercion
```

```
Cldat <- as.data.frame(Cldat)
head(Cldat)
```

```
##   Sample_ID Cl_ppm Cl_Area
## 1 Lab Blank    NA     NA
## 2 Lab Blank    NA     NA
## 3 Lab Blank    NA     NA
## 4 Standard 1  5.7300  8.0833
## 5 Standard 2 11.3295 15.9825
## 6 Standard 3 20.3441 28.6992
```

```
## Bring the data back together:
```

```
all_dat <- merge(Sdat, Cldat, by="Sample_ID")
head(all_dat)
```

```
##   Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area
## 1          707.6693  23.7212 4345.0138 286.6714
## 2          707.6693  23.7212 1400.2957  69.8002
## 3          707.6693  23.7212   0.0058   0.0082
## 4          707.6693  23.7212 1605.3399  61.2697
## 5          707.6693  23.7212        NA        NA
## 6          189.0118   8.5230 4345.0138 286.6714
```

```
## Remove empty lines
```

```
all_dat <- all_dat[!(is.na(all_dat$Sample_ID) | all_dat$Sample_ID==""), ]
head(all_dat)
```

```
##   Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area
## 26 Lab Blank    NA     NA    NA     NA
## 27 Lab Blank    NA     NA    NA     NA
## 28 Lab Blank    NA     NA    NA     NA
## 29 Lab Blank    NA     NA    NA     NA
## 30 Lab Blank    NA     NA  0.0196  0.0277
## 31 Lab Blank    NA     NA  0.0100  0.0141
```

```
all_dat <- droplevels(all_dat[!all_dat$Sample_ID == 'Lab Blank',])
head(all_dat)
```

```
##   Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area
## 170 Standard 1  0.4565   0.5324  5.7300  8.0833
## 171 Standard 1  0.4565   0.5324  5.8701  8.2810
## 172 Standard 1  0.4565   0.5324  5.7836  8.1589
## 173 Standard 1  0.4826   0.5628  5.7300  8.0833
## 174 Standard 1  0.4826   0.5628  5.8701  8.2810
## 175 Standard 1  0.4826   0.5628  5.7836  8.1589
```

```
all_dat[is.na(all_dat)] <- 0
```

Pull out standards - could do some checks here if we want

```
stds <- all_dat[grepl("Standard", all_dat$Sample_ID),]  
head(stds)
```

```
##      Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area  
## 170 Standard 1  0.4565  0.5324 5.7300 8.0833  
## 171 Standard 1  0.4565  0.5324 5.8701 8.2810  
## 172 Standard 1  0.4565  0.5324 5.7836 8.1589  
## 173 Standard 1  0.4826  0.5628 5.7300 8.0833  
## 174 Standard 1  0.4826  0.5628 5.8701 8.2810  
## 175 Standard 1  0.4826  0.5628 5.7836 8.1589
```

```
stds_chk_S <- stds %>%  
  group_by(Sample_ID) %>%  
  summarise(mean = mean(S04_ppm), sd = sd(S04_ppm))  
stds_chk_S$cv <- (stds_chk_S$sd/stds_chk_S$mean)*100  
stds_chk_S$flag <- ifelse(stds_chk_S$cv < 2, 'YES', 'NO, rerun')  
head(stds_chk_S)
```

```
## # A tibble: 5 x 5  
##   Sample_ID   mean     sd    cv flag  
##   <fct>     <dbl>  <dbl> <dbl> <chr>  
## 1 Standard 1  0.472 0.0118 2.50 NO, rerun  
## 2 Standard 2  0.985 0.0218 2.21 NO, rerun  
## 3 Standard 3  1.92  0.00999 0.520 YES  
## 4 Standard 4 10.1  0.121  1.21 YES  
## 5 Standard 5 20.2  0.183  0.904 YES
```

```
stds_chk_Cl <- stds %>%  
  group_by(Sample_ID) %>%  
  summarise(mean = mean(Cl_ppm), sd = sd(Cl_ppm))  
stds_chk_Cl$cv <- (stds_chk_Cl$sd/stds_chk_Cl$mean)*100  
stds_chk_Cl$flag <- ifelse(stds_chk_Cl$cv < 2, 'YES', 'NO, rerun')  
head(stds_chk_Cl)
```

```
## # A tibble: 5 x 5  
##   Sample_ID   mean     sd    cv flag  
##   <fct>     <dbl>  <dbl> <dbl> <chr>  
## 1 Standard 1   5.79 0.0612 1.06 YES  
## 2 Standard 2  11.4 0.0937 0.819 YES  
## 3 Standard 3  20.5 0.159  0.777 YES  
## 4 Standard 4 101.  1.14  1.12 YES  
## 5 Standard 5 201.  1.98  0.982 YES
```

```
lmS <- lm(stds$SO4_Area ~ stds$SO4_ppm)
SO4_sum <- summary(lmS)
SO4_Slope <- SO4_sum$coefficients[2, 1]
SO4_Int <- SO4_sum$coefficients[1, 1]

lmCl <- lm(stds$Cl_Area ~ stds$Cl_ppm)
Cl_sum <- summary(lmCl)
Cl_Slope <- Cl_sum$coefficients[2, 1]
Cl_Int <- Cl_sum$coefficients[1, 1]
```

Calculate mmol/L concentrations

```
#remove standards from sample dataframe
sampledat <- all_dat[!grepl("Standard", all_dat$Sample_ID),]
head(sampledat)
```

```
##                               Sample_ID  SO4_ppm SO4_Area   Cl_ppm  Cl_Area
## 205 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849  13.5575 4335.7195 122.3274
## 206 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306  12.7632 4145.0144 116.9468
## 207 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434  12.9198 4153.6146 117.1895
## 208          TEMPEST_FW_B4_20230607_0800  11.1494  11.8200   27.9934  35.9001
## 209          TEMPEST_FW_B4_20230607_1600   2.4624   2.2973  206.4674 233.0096
## 210          TEMPEST_FW_C3_20230607_0800  10.7750  11.4231   23.9976  30.7756
```

```
# Constants needed for calculations:
clmw <- 35.45      #molecular weight of Chloride: 35.45
smw <- 32.06      #molecular weight of sulfur: 32.06

# Convert ppm to mmol/L
sampledat$SO4_mM <- (sampledat$SO4_ppm / smw)
sampledat$Cl_mM <- (sampledat$Cl_ppm / clmw)

# Calculate Salinity
# calculated using the Knudsen equation
# Salinity = 0.03 + 1.8050 * Chlorinity
# Ref: A Practical Handbook of Seawater Analysis by Strickland & Parsons (P. 11)
# =((1.807*Cl_ppm)+0.026)/1000
sampledat$salinity <- ((1.8070 * sampledat$Cl_ppm) + 0.026) / 1000

head(sampledat)
```

```
##                               Sample_ID  SO4_ppm SO4_Area   Cl_ppm  Cl_Area
## 205 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849  13.5575 4335.7195 122.3274
## 206 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306  12.7632 4145.0144 116.9468
## 207 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434  12.9198 4153.6146 117.1895
## 208          TEMPEST_FW_B4_20230607_0800  11.1494  11.8200   27.9934  35.9001
```

```
## 209          TEMPEST_FW_B4_20230607_1600    2.4624    2.2973    206.4674 233.0096
## 210          TEMPEST_FW_C3_20230607_0800   10.7750   11.4231    23.9976 30.7756
##          S04_mM          Cl_mM    salinity
## 205 18.13115721 122.3052045 7.83467114
## 206 17.06895197 116.9256530 7.49006702
## 207 17.27833437 117.1682539 7.50560758
## 208 0.34776669 0.7896587 0.05061007
## 209 0.07680599 5.8241862 0.37311259
## 210 0.33608858 0.6769422 0.04338966
```

Pull out dups and check with percent difference

```
#Show me the data that we have from the calculations
head(sampledat)
```

```
##          Sample_ID S04_ppm S04_Area    Cl_ppm Cl_Area
## 205 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849 13.5575 4335.7195 122.3274
## 206 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306 12.7632 4145.0144 116.9468
## 207 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434 12.9198 4153.6146 117.1895
## 208          TEMPEST_FW_B4_20230607_0800 11.1494 11.8200 27.9934 35.9001
## 209          TEMPEST_FW_B4_20230607_1600 2.4624 2.2973 206.4674 233.0096
## 210          TEMPEST_FW_C3_20230607_0800 10.7750 11.4231 23.9976 30.7756
##          S04_mM          Cl_mM    salinity
## 205 18.13115721 122.3052045 7.83467114
## 206 17.06895197 116.9256530 7.49006702
## 207 17.27833437 117.1682539 7.50560758
## 208 0.34776669 0.7896587 0.05061007
## 209 0.07680599 5.8241862 0.37311259
## 210 0.33608858 0.6769422 0.04338966
```

```
#pull out any rows that have "dup" in the SampleID column
dups <- sampledat %>%
  filter(str_detect(Sample_ID, "_dup")) #have to change this to match data
head(dups)
```

```
##          Sample_ID S04_ppm S04_Area    Cl_ppm Cl_Area
## 1 TEMPEST_FW_SOURCE_20230607_1519_dup 12.6336 13.3935 0.000 0.0000
## 2          TEMPEST_SW_B4_20230607_0800_dup 403.8132 9.4182 3303.890 93.2155
## 3          TEMPEST_SW_B4_20230607_1600_dup 440.2724 10.2686 3510.899 99.0560
## 4          TEMPEST_SW_D5_20230607_0800_dup 292.2655 6.8166 2343.111 66.1082
## 5          TEMPEST_SW_I5_20230607_0800_dup 273.6440 6.3823 2744.173 77.4237
##          S04_mM          Cl_mM salinity
## 1 0.3940611 0.00000 0.000026
## 2 12.5955459 93.19858 5.970155
## 3 13.7327636 99.03805 6.344220
## 4 9.1162040 66.09623 4.234028
## 5 8.5353712 77.40967 4.958746
```

```
#remove these from sample dataframe in a new dataframe
sampledat2 <- sampledat %>%
  filter(!str_detect(Sample_ID, "_dup")) %>%
```

```
filter(!str_detect(Sample_ID, "_spk"))
head(sampledat2)
```

```
##
##           Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area
## 1 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849 13.5575 4335.7195 122.3274
## 2 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306 12.7632 4145.0144 116.9468
## 3 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434 12.9198 4153.6146 117.1895
## 4 TEMPEST_FW_B4_20230607_0800 11.1494 11.8200 27.9934 35.9001
## 5 TEMPEST_FW_B4_20230607_1600 2.4624 2.2973 206.4674 233.0096
## 6 TEMPEST_FW_C3_20230607_0800 10.7750 11.4231 23.9976 30.7756
##           S04_mM Cl_mM salinity
## 1 18.13115721 122.3052045 7.83467114
## 2 17.06895197 116.9256530 7.49006702
## 3 17.27833437 117.1682539 7.50560758
## 4 0.34776669 0.7896587 0.05061007
## 5 0.07680599 5.8241862 0.37311259
## 6 0.33608858 0.6769422 0.04338966
```

```
#remove the dup from these IDs so we will have duplicate sample names
dups$Sample_ID<-gsub("_dup","",as.character(dups$Sample_ID))
dups <- dups[ , -c(2:5)]
colnames(dups) <- c('Sample_ID', 'S04_mM_dup', 'Cl_mM_dup', 'salinity_dup')
head(dups)
```

```
##
##           Sample_ID S04_mM_dup Cl_mM_dup salinity_dup
## 1 TEMPEST_FW_SOURCE_20230607_1519 0.3940611 0.00000 0.000026
## 2 TEMPEST_SW_B4_20230607_0800 12.5955459 93.19858 5.970155
## 3 TEMPEST_SW_B4_20230607_1600 13.7327636 99.03805 6.344220
## 4 TEMPEST_SW_D5_20230607_0800 9.1162040 66.09623 4.234028
## 5 TEMPEST_SW_I5_20230607_0800 8.5353712 77.40967 4.958746
```

```
#put it back together with the old data set and look for duplicates
QAdups <- merge(sampledat2, dups)
head(QAdups)
```

```
##
##           Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area S04_mM
## 1 TEMPEST_FW_SOURCE_20230607_1519 12.7863 13.5553 0.000 0.0000 0.3988241
## 2 TEMPEST_SW_B4_20230607_0800 377.6524 8.8081 3087.801 87.1188 11.7795508
## 3 TEMPEST_SW_B4_20230607_1600 440.3089 10.2694 3507.065 98.9478 13.7339021
## 4 TEMPEST_SW_D5_20230607_0800 293.4522 6.8443 2353.532 66.4022 9.1532190
## 5 TEMPEST_SW_I5_20230607_0800 284.5868 6.6375 2842.504 80.1980 8.8766937
##           Cl_mM salinity S04_mM_dup Cl_mM_dup salinity_dup
## 1 0.00000 0.000026 0.3940611 0.00000 0.000026
## 2 87.10298 5.579682 12.5955459 93.19858 5.970155
## 3 98.92990 6.337292 13.7327636 99.03805 6.344220
## 4 66.39019 4.252859 9.1162040 66.09623 4.234028
## 5 80.18346 5.136430 8.5353712 77.40967 4.958746
```

```
QAdups$S04_dups_chk <- ((abs(QAdups$S04_mM-QAdups$S04_mM_dup))/((QAdups$S04_mM+QAdups$S04_mM_dup)/2))*10
QAdups$S04_dups_flag <- ifelse(QAdups$S04_dups_chk <10, 'YES', 'NO, rerun')
```

```
QAdups$Cl_dups_chk <- ((abs(QAdups$Cl_mM-QAdups$Cl_mM_dup))/((QAdups$Cl_mM+QAdups$Cl_mM_dup)/2))*100
QAdups$Cl_dups_flag <- ifelse(QAdups$Cl_dups_chk <10, 'YES', 'NO, rerun')

head(QAdups)
```

```
##           Sample_ID  S04_ppm S04_Area  Cl_ppm Cl_Area  S04_mM
## 1 TEMPEST_FW_SOURCE_20230607_1519 12.7863 13.5553 0.000 0.0000 0.3988241
## 2 TEMPEST_SW_B4_20230607_0800 377.6524 8.8081 3087.801 87.1188 11.7795508
## 3 TEMPEST_SW_B4_20230607_1600 440.3089 10.2694 3507.065 98.9478 13.7339021
## 4 TEMPEST_SW_D5_20230607_0800 293.4522 6.8443 2353.532 66.4022 9.1532190
## 5 TEMPEST_SW_I5_20230607_0800 284.5868 6.6375 2842.504 80.1980 8.8766937
##           Cl_mM salinity S04_mM_dup Cl_mM_dup salinity_dup S04_dups_chk
## 1 0.00000 0.000026 0.3940611 0.00000 0.000026 1.201420934
## 2 87.10298 5.579682 12.5955459 93.19858 5.970155 6.695317107
## 3 98.92990 6.337292 13.7327636 99.03805 6.344220 0.008289978
## 4 66.39019 4.252859 9.1162040 66.09623 4.234028 0.405212272
## 5 80.18346 5.136430 8.5353712 77.40967 4.958746 3.920528928
## S04_dups_flag Cl_dups_chk Cl_dups_flag
## 1 YES NaN <NA>
## 2 YES 6.7615540 YES
## 3 YES 0.1092568 YES
## 4 YES 0.4437552 YES
## 5 YES 3.5201930 YES
```

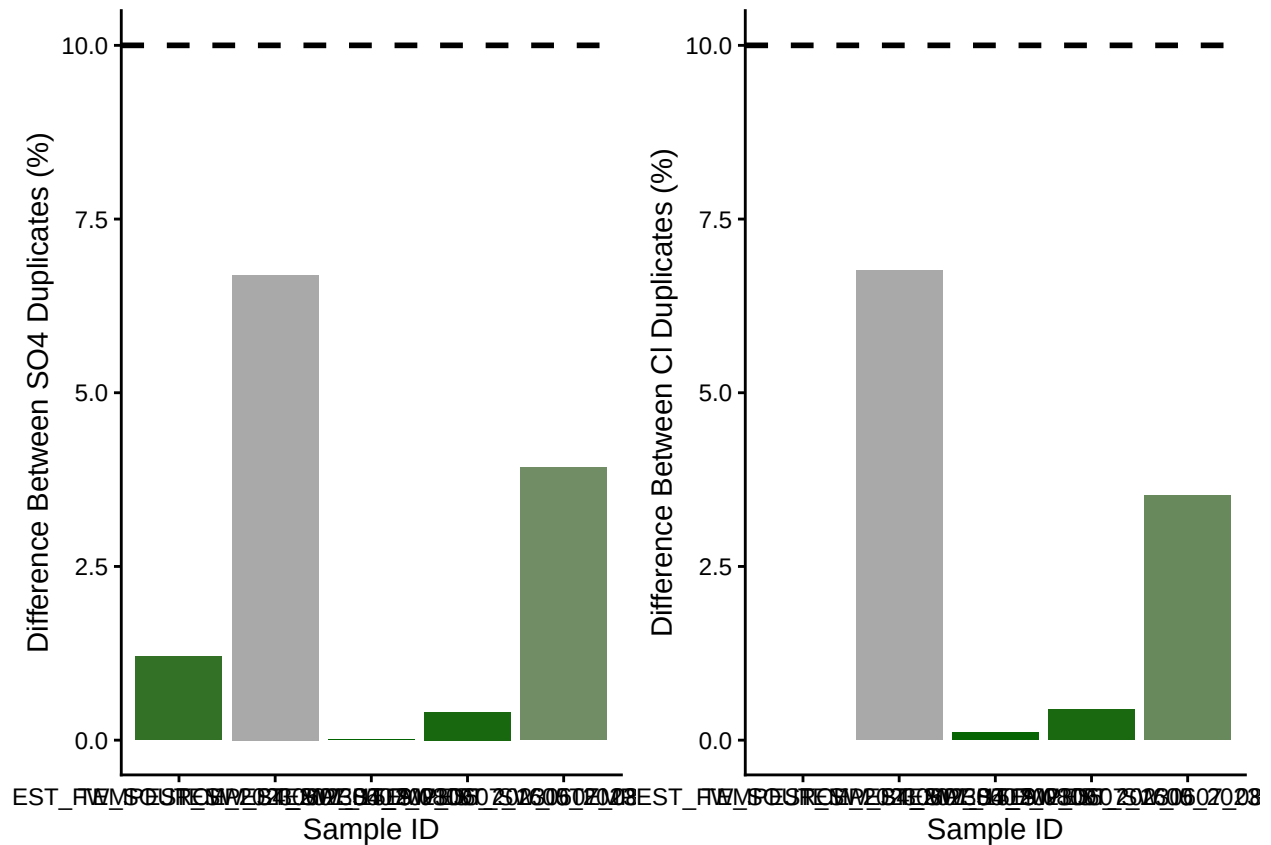
```
#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_chk, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="Difference Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar, ncol=2, nrow=1)
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').
```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_dups <- QAdups %>%
  group_by(S04_dups_flag) %>%
  summarise(S_no_rows = length(S04_dups_flag))
Perc_dups1 <- QAdups %>%
  group_by(CI_dups_flag) %>%
  summarise(CI_no_rows = length(CI_dups_flag))
colnames(Perc_dups) <- c("Flag", "S_no_rows")
colnames(Perc_dups1) <- c("Flag", "CI_no_rows")
Perc_dups2 <- cbind(Perc_dups, Perc_dups1)

Perc_dups2$Total <- length(QAdups$S04_dups_flag)
Perc_dups2$S_Percent <- (Perc_dups2$S_no_rows / Perc_dups2$Total)*100
Perc_dups2$CI_Percent <- (Perc_dups2$CI_no_rows / Perc_dups2$Total)*100
head(Perc_dups2)
```

```
##   Flag S_no_rows Flag CI_no_rows Total S_Percent CI_Percent
## 1  YES         5  YES         4     5      100         80
## 2  YES         5 <NA>         1     5      100         20
```

Pull out dups and check with cv

```

#the cv
# calculate the sd bewteen the two duplicates then divide by the mean and multiply by 100

df2 <- as.data.frame(QAdups$S04_mM)
df2$dups <- QAdups$S04_mM_dup

df2$sds <- apply(df2,1,sd)

QAdups$S04_dups_cv <- (df2$sds)/((QAdups$S04_mM+QAdups$S04_mM_dup)/2) * 100
QAdups$S04_dups_cv_flag <- ifelse(QAdups$S04_dups_cv <11, 'YES', 'NO, rerun')

head(QAdups)

```

```

##           Sample_ID S04_ppm S04_Area Cl_ppm Cl_Area S04_mM
## 1 TEMPEST_FW_SOURCE_20230607_1519 12.7863 13.5553 0.000 0.0000 0.3988241
## 2 TEMPEST_SW_B4_20230607_0800 377.6524 8.8081 3087.801 87.1188 11.7795508
## 3 TEMPEST_SW_B4_20230607_1600 440.3089 10.2694 3507.065 98.9478 13.7339021
## 4 TEMPEST_SW_D5_20230607_0800 293.4522 6.8443 2353.532 66.4022 9.1532190
## 5 TEMPEST_SW_I5_20230607_0800 284.5868 6.6375 2842.504 80.1980 8.8766937
## Cl_mM salinity S04_mM_dup Cl_mM_dup salinity_dup S04_dups_chk
## 1 0.00000 0.000026 0.3940611 0.00000 0.000026 1.201420934
## 2 87.10298 5.579682 12.5955459 93.19858 5.970155 6.695317107
## 3 98.92990 6.337292 13.7327636 99.03805 6.344220 0.008289978
## 4 66.39019 4.252859 9.1162040 66.09623 4.234028 0.405212272
## 5 80.18346 5.136430 8.5353712 77.40967 4.958746 3.920528928
## S04_dups_flag Cl_dups_chk Cl_dups_flag S04_dups_cv S04_dups_cv_flag
## 1 YES NaN <NA> 0.8495329 YES
## 2 YES 6.7615540 YES 4.7343041 YES
## 3 YES 0.1092568 YES 0.0058619 YES
## 4 YES 0.4437552 YES 0.2865283 YES
## 5 YES 3.5201930 YES 2.7722326 YES

```

```

#plot dups output as a bar graph to easily check - want any over 10% to be red need to work on this
Sdupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = S04_dups_cv, fill=S04_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between S04 Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

Cldupsbar <- ggplot(data = QAdups, aes(x = Sample_ID, y = Cl_dups_chk, fill=Cl_dups_chk)) +
  geom_bar(stat = 'identity') +
  scale_fill_gradient2(low='darkgreen', mid='darkgreen', high='darkgrey', space='Lab') +
  theme_classic() + labs(x= "Sample ID", y="CV Between Cl Duplicates (%)") +
  theme(legend.position="none") + geom_hline(yintercept=10, linetype="dashed",
  color = "black", size=1)

ggarrange(Sdupsbar, Cldupsbar,ncol=2, nrow=1)

```

```

## Warning: Removed 1 row containing missing values or values outside the scale range
## ('geom_bar()').

```



```
##Merge sample data with dilution data
Dilutions <- read.csv("Raw Data/TMP 2023 Sample Dilutions from Dionex.csv")
sampledat_dils <- merge(sampledat, Dilutions, by = "Sample_ID")
unique(sampledat_dils$Dilution)

## [1] 50.00  1.10  1.25 56.51 52.74

#Show me the data that we have from the calculations
head(sampledat_dils)

##              Sample_ID  S04_ppm S04_Area   Cl_ppm  Cl_Area
## 1 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849 13.5575 4335.7195 122.3274
## 2 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306 12.7632 4145.0144 116.9468
## 3 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434 12.9198 4153.6146 117.1895
## 4      TEMPEST_FW_B4_20230607_0800 11.1494 11.8200 27.9934 35.9001
## 5      TEMPEST_FW_B4_20230607_1600 2.4624 2.2973 206.4674 233.0096
## 6      TEMPEST_FW_C3_20230607_0800 10.7750 11.4231 23.9976 30.7756
##      S04_mM      Cl_mM  salinity Dilution
## 1 18.13115721 122.3052045 7.83467114 50.00
## 2 17.06895197 116.9256530 7.49006702 50.00
## 3 17.27833437 117.1682539 7.50560758 50.00
## 4 0.34776669 0.7896587 0.05061007 1.10
## 5 0.07680599 5.8241862 0.37311259 1.25
## 6 0.33608858 0.6769422 0.04338966 1.10

#pull out any rows that have "spk" in the SampleID column
spks <- sampledat_dils %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)

##              Sample_ID  S04_ppm S04_Area   Cl_ppm  Cl_Area
## 1 TEMPEST_FW_SOURCE_20230607_0611_spk 16.0473 17.0125 23.1723 29.7173
## 2      TEMPEST_SW_E3_20230607_1600_spk 332.1500 7.7468 2355.1999 66.4493
## 3      TEMPEST_SW_I5_20230607_1600_spk 468.0219 10.9158 3179.1880 89.6971
## 4 TEMPEST_SW_SOURCE_20230607_0621_spk 563.1599 13.1347 4002.2094 112.9177
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 707.6693 16.5051 4345.0138 122.5896
##      S04_mM      Cl_mM  salinity Dilution
## 1 0.5005396 0.6536615 0.04189835 1.1
## 2 10.3602620 66.4372327 4.25587222 50.0
## 3 14.5983125 89.6809027 5.74481872 50.0
## 4 17.5658110 112.8973032 7.23201839 50.0
## 5 22.0732782 122.5673850 7.85146594 50.0

#remove the dup from these IDs so we will have duplicate sample names
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
spks <- spks[,c(1,6)]
colnames(spks) <- c('Sample_ID', 'S04_mM_spk')
head(spks)

##              Sample_ID S04_mM_spk
```

```
## 1    TEMPEST_FW_SOURCE_20230607_0611  0.5005396
## 2      TEMPEST_SW_E3_20230607_1600 10.3602620
## 3      TEMPEST_SW_I5_20230607_1600 14.5983125
## 4    TEMPEST_SW_SOURCE_20230607_0621 17.5658110
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 22.0732782
```

```
#put it back together with the old data set and look for duplicates
QAspks <- merge(sampladat_dils, spks)
head(QAspks)
```

```
##              Sample_ID  SO4_ppm SO4_Area   Cl_ppm  Cl_Area
## 1    TEMPEST_FW_SOURCE_20230607_0611  12.7760  13.5445   22.8178  29.2627
## 2      TEMPEST_SW_E3_20230607_1600 259.0745   6.0425 2327.3879  65.6646
## 3      TEMPEST_SW_I5_20230607_1600 336.2264   7.8419 3120.4635  88.0403
## 4    TEMPEST_SW_SOURCE_20230607_0621 571.0464  13.3187 4300.3781 121.3302
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 707.6693  16.5051 4345.0138 122.5896
##      SO4_mM      Cl_mM  salinity Dilution SO4_mM_spk
## 1  0.3985028  0.6436615 0.04125776      1.1  0.5005396
## 2  8.0809264  65.6526911 4.20561594     50.0 10.3602620
## 3 10.4874111  88.0243583 5.63870354     50.0 14.5983125
## 4 17.8118029 121.3082680 7.77080923     50.0 17.5658110
## 5 22.0732782 122.5673850 7.85146594     50.0 22.0732782
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
spkconc <- (250/smw)      # in mM
spkvol <- 10              # in uL
spkvol <- spkvol/1000000
#spike for these samples was 10uL of the 250mM standard
QAspks$SO4_spk_Conc <- (spkconc)*spkvol      # mmoles of SO4
head(QAspks)
```

```
##              Sample_ID  SO4_ppm SO4_Area   Cl_ppm  Cl_Area
## 1    TEMPEST_FW_SOURCE_20230607_0611  12.7760  13.5445   22.8178  29.2627
## 2      TEMPEST_SW_E3_20230607_1600 259.0745   6.0425 2327.3879  65.6646
## 3      TEMPEST_SW_I5_20230607_1600 336.2264   7.8419 3120.4635  88.0403
## 4    TEMPEST_SW_SOURCE_20230607_0621 571.0464  13.3187 4300.3781 121.3302
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 707.6693  16.5051 4345.0138 122.5896
##      SO4_mM      Cl_mM  salinity Dilution SO4_mM_spk SO4_spk_Conc
## 1  0.3985028  0.6436615 0.04125776      1.1  0.5005396 7.797879e-05
## 2  8.0809264  65.6526911 4.20561594     50.0 10.3602620 7.797879e-05
## 3 10.4874111  88.0243583 5.63870354     50.0 14.5983125 7.797879e-05
## 4 17.8118029 121.3082680 7.77080923     50.0 17.5658110 7.797879e-05
## 5 22.0732782 122.5673850 7.85146594     50.0 22.0732782 7.797879e-05
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
# QAspks$Dilution <- 1
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "TEMPEST"), 50, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 50, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 50, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 100, QAspks$Dilution)
```

```

# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 50, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 50, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 100, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 100, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 100, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 200, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$Dilution)
# QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$Dilution)

#Set Sample volumes in uL
unique(QAspks$Dilution)

```

```
## [1] 1.1 50.0
```

```

QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(QAspks$Dilution == 50, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 56.85, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 51.16, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 49.27, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 56.51, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 5.15, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 1.25, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 1.1, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 52.74, 1483, QAspks$SampleVol)

# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "TEMPEST"), 1483, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 1462, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000000
head(QAspks)

```

```

##           Sample_ID  SO4_ppm SO4_Area   Cl_ppm  Cl_Area
## 1  TEMPEST_FW_SOURCE_20230607_0611  12.7760  13.5445   22.8178  29.2627
## 2      TEMPEST_SW_E3_20230607_1600  259.0745   6.0425  2327.3879  65.6646
## 3      TEMPEST_SW_I5_20230607_1600  336.2264   7.8419  3120.4635  88.0403
## 4  TEMPEST_SW_SOURCE_20230607_0621  571.0464  13.3187  4300.3781 121.3302
## 5 TEMPEST_SW_SOURCE_20230607_1515spk  707.6693  16.5051  4345.0138 122.5896
##      SO4_mM      Cl_mM  salinity Dilution SO4_mM_spk SO4_spk_Conc SampleVol
## 1  0.3985028  0.6436615 0.04125776      1.1  0.5005396 7.797879e-05  0.001500

```

```
## 2 8.0809264 65.6526911 4.20561594 50.0 10.3602620 7.797879e-05 0.001483
## 3 10.4874111 88.0243583 5.63870354 50.0 14.5983125 7.797879e-05 0.001483
## 4 17.8118029 121.3082680 7.77080923 50.0 17.5658110 7.797879e-05 0.001483
## 5 22.0732782 122.5673850 7.85146594 50.0 22.0732782 7.797879e-05 0.001483
```

```
#gives us the total SO4 in the sample in mmoles
```

```
QAspks$SO4_Total_unspkd <- (QAspks$SO4_mM/QAspks$Dilution)*(QAspks$SampleVol)
```

```
##total SO4 in spiked sample in mmoles
```

```
QAspks$SO4_Total_spkd <- (QAspks$SO4_mM_spk/QAspks$Dilution)*(QAspks$SampleVol+spkvol)
```

```
QAspks$SO4_expctd_spkd <- (QAspks$SO4_Total_unspkd + QAspks$SO4_spk_Conc)
```

```
QAspks$spk_recovery <- (QAspks$SO4_Total_spkd/QAspks$SO4_expctd_spkd)*100
```

```
QAspks$SO4_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO, rerun')
```

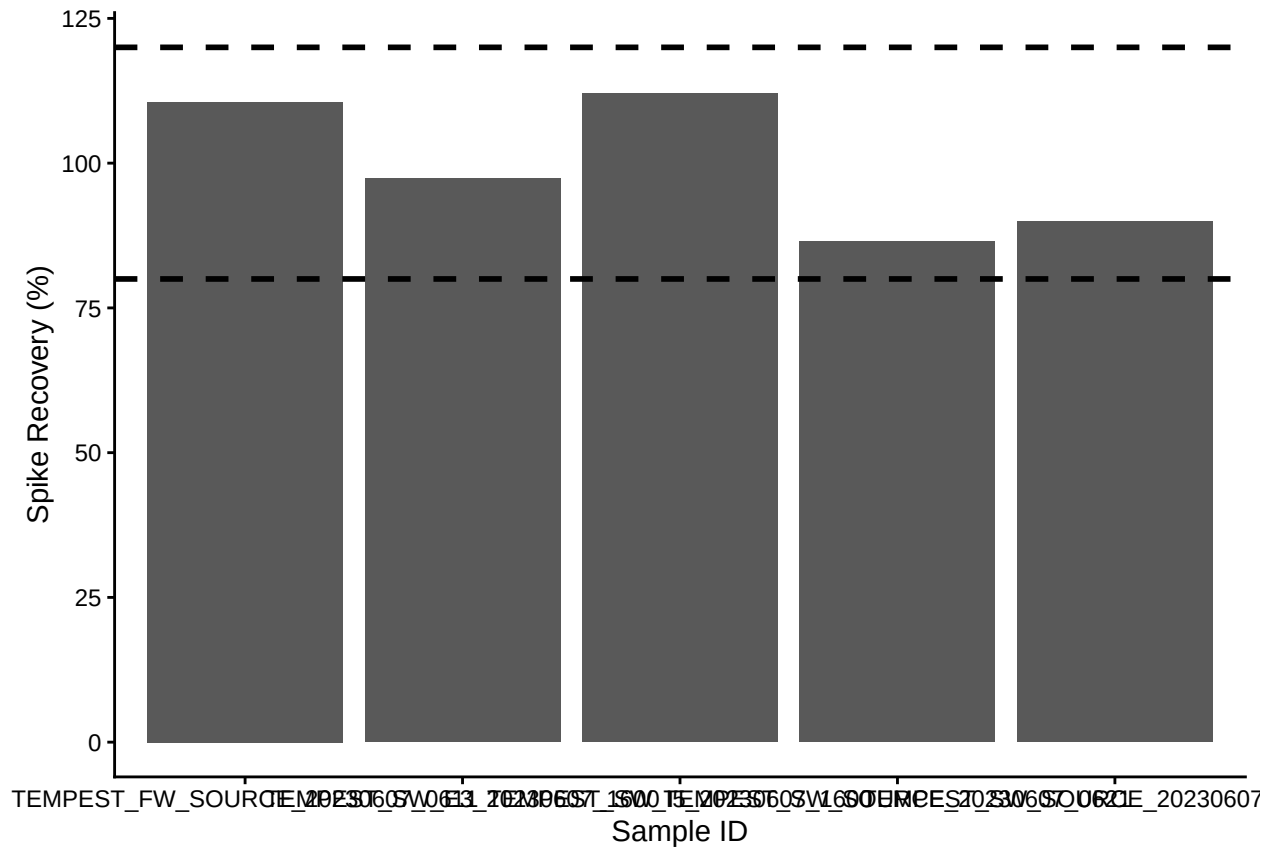
```
head(QAspks)
```

```
##           Sample_ID  SO4_ppm SO4_Area  Cl_ppm  Cl_Area
## 1 TEMPEST_FW_SOURCE_20230607_0611 12.7760 13.5445 22.8178 29.2627
## 2 TEMPEST_SW_E3_20230607_1600 259.0745 6.0425 2327.3879 65.6646
## 3 TEMPEST_SW_I5_20230607_1600 336.2264 7.8419 3120.4635 88.0403
## 4 TEMPEST_SW_SOURCE_20230607_0621 571.0464 13.3187 4300.3781 121.3302
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 707.6693 16.5051 4345.0138 122.5896
##      SO4_mM      Cl_mM  salinity Dilution SO4_mM_spk SO4_spk_Conc SampleVol
## 1 0.3985028 0.6436615 0.04125776 1.1 0.5005396 7.797879e-05 0.001500
## 2 8.0809264 65.6526911 4.20561594 50.0 10.3602620 7.797879e-05 0.001483
## 3 10.4874111 88.0243583 5.63870354 50.0 14.5983125 7.797879e-05 0.001483
## 4 17.8118029 121.3082680 7.77080923 50.0 17.5658110 7.797879e-05 0.001483
## 5 22.0732782 122.5673850 7.85146594 50.0 22.0732782 7.797879e-05 0.001483
## SO4_Total_unspkd SO4_Total_spkd SO4_expctd_spkd spk_recovery SO4_spks_flag
## 1 0.0005434129 0.0006871044 0.0006213917 110.57508 YES
## 2 0.0002396803 0.0003093574 0.0003176591 97.38662 YES
## 3 0.0003110566 0.0004359056 0.0003890354 112.04780 YES
## 4 0.0005282981 0.0005245151 0.0006062769 86.51412 YES
## 5 0.0006546934 0.0006591081 0.0007326722 89.95948 YES
```

```
#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on
```

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

```
spksbar
```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 1 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 YES             5     5    100
```

Pull out spikes and check with area calc

```
##Merge sample data with dilution data
Dilutions <- read.csv("Raw Data/TMP 2023 Sample Dilutions from Dionex.csv")
sampledat_dils <- merge(sampledat, Dilutions, by = "Sample_ID")
unique(sampledat_dils$Dilution)
```

```
## [1] 50.00  1.10  1.25 56.51 52.74
```

```
#Show me the data that we have from the calculations
head(sampledat_dils)
```

```
##
##      Sample_ID  S04_ppm S04_Area  Cl_ppm Cl_Area
## 1 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849 13.5575 4335.7195 122.3274
## 2 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306 12.7632 4145.0144 116.9468
## 3 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434 12.9198 4153.6146 117.1895
## 4      TEMPEST_FW_B4_20230607_0800 11.1494 11.8200 27.9934 35.9001
## 5      TEMPEST_FW_B4_20230607_1600 2.4624 2.2973 206.4674 233.0096
## 6      TEMPEST_FW_C3_20230607_0800 10.7750 11.4231 23.9976 30.7756
##      S04_mM      Cl_mM      salinity Dilution
## 1 18.13115721 122.3052045 7.83467114 50.00
## 2 17.06895197 116.9256530 7.49006702 50.00
## 3 17.27833437 117.1682539 7.50560758 50.00
## 4 0.34776669 0.7896587 0.05061007 1.10
## 5 0.07680599 5.8241862 0.37311259 1.25
## 6 0.33608858 0.6769422 0.04338966 1.10
```

```
#now use area of all samples and calculate undilution corrected concentrations in ug/mL
sampledat_dils$S04_ugmL <- ((sampledat_dils$S04_Area)-S04_Int)/S04_Slope
sampledat_dils$Cl_ugmL <- (sampledat_dils$Cl_Area-Cl_Int)/Cl_Slope
head(sampledat_dils)
```

```
##
##      Sample_ID  S04_ppm S04_Area  Cl_ppm Cl_Area
## 1 TEMPEST_ESTUARY_SOURCE_20230607_0710 581.2849 13.5575 4335.7195 122.3274
## 2 TEMPEST_ESTUARY_SOURCE_20230607_1346 547.2306 12.7632 4145.0144 116.9468
## 3 TEMPEST_ESTUARY_SOURCE_20230607_1700 553.9434 12.9198 4153.6146 117.1895
## 4      TEMPEST_FW_B4_20230607_0800 11.1494 11.8200 27.9934 35.9001
## 5      TEMPEST_FW_B4_20230607_1600 2.4624 2.2973 206.4674 233.0096
## 6      TEMPEST_FW_C3_20230607_0800 10.7750 11.4231 23.9976 30.7756
##      S04_mM      Cl_mM      salinity Dilution S04_ugmL Cl_ugmL
## 1 18.13115721 122.3052045 7.83467114 50.00 11.625713 86.71442
## 2 17.06895197 116.9256530 7.49006702 50.00 10.944591 82.90027
## 3 17.27833437 117.1682539 7.50560758 50.00 11.078877 83.07231
## 4 0.34776669 0.7896587 0.05061007 1.10 10.135786 25.44856
## 5 0.07680599 5.8241862 0.37311259 1.25 1.969955 165.17390
## 6 0.33608858 0.6769422 0.04338966 1.10 9.795439 21.81594
```

```
#pull out any rows that have "spk" in the SampleID column
spks <- sampledat_dils %>%
  filter(str_detect(Sample_ID, "spk")) #have to change this to match data
head(spks)
```

```
##
##      Sample_ID  S04_ppm S04_Area  Cl_ppm Cl_Area
## 1 TEMPEST_FW_SOURCE_20230607_0611_spk 16.0473 17.0125 23.1723 29.7173
## 2      TEMPEST_SW_E3_20230607_1600_spk 332.1500 7.7468 2355.1999 66.4493
## 3      TEMPEST_SW_I5_20230607_1600_spk 468.0219 10.9158 3179.1880 89.6971
## 4 TEMPEST_SW_SOURCE_20230607_0621_spk 563.1599 13.1347 4002.2094 112.9177
## 5 TEMPEST_SW_SOURCE_20230607_1515spk 707.6693 16.5051 4345.0138 122.5896
##      S04_mM      Cl_mM      salinity Dilution S04_ugmL Cl_ugmL
## 1 0.5005396 0.6536615 0.04189835 1.1 14.588417 21.06574
## 2 10.3602620 66.4372327 4.25587222 50.0 6.642967 47.10402
```

```
## 3 14.5983125 89.6809027 5.74481872 50.0 9.360423 63.58373
## 4 17.5658110 112.8973032 7.23201839 50.0 11.263157 80.04415
## 5 22.0732782 122.5673850 7.85146594 50.0 14.153315 86.90029
```

```
## Only keep the columns that we need
```

```
spks <- spks[,c(1,9,10)]
head(spks)
```

```
##                               Sample_ID Dilution  S04_ugmL
## 1 TEMPEST_FW_SOURCE_20230607_0611_spk      1.1 14.588417
## 2      TEMPEST_SW_E3_20230607_1600_spk     50.0  6.642967
## 3      TEMPEST_SW_I5_20230607_1600_spk     50.0  9.360423
## 4 TEMPEST_SW_SOURCE_20230607_0621_spk     50.0 11.263157
## 5  TEMPEST_SW_SOURCE_20230607_1515spk     50.0 14.153315
```

```
#remove the spk from these IDs so we will have duplicate sample names
```

```
spks$Sample_ID<-gsub("_spk","",as.character(spks$Sample_ID))
colnames(spks) <- c('Sample_ID', 'Dilution', 'S04_ugmL_spk')
head(spks)
```

```
##                               Sample_ID Dilution S04_ugmL_spk
## 1      TEMPEST_FW_SOURCE_20230607_0611      1.1 14.588417
## 2      TEMPEST_SW_E3_20230607_1600      50.0  6.642967
## 3      TEMPEST_SW_I5_20230607_1600      50.0  9.360423
## 4      TEMPEST_SW_SOURCE_20230607_0621      50.0 11.263157
## 5 TEMPEST_SW_SOURCE_20230607_1515spk      50.0 14.153315
```

```
#put it back together with the old data set and look for duplicates
```

```
QAspks <- merge(sampledat_dils, spks)
head(QAspks)
```

```
##                               Sample_ID Dilution  S04_ppm S04_Area  Cl_ppm
## 1      TEMPEST_FW_SOURCE_20230607_0611      1.1 12.7760 13.5445 22.8178
## 2      TEMPEST_SW_E3_20230607_1600      50.0 259.0745  6.0425 2327.3879
## 3      TEMPEST_SW_I5_20230607_1600      50.0 336.2264  7.8419 3120.4635
## 4      TEMPEST_SW_SOURCE_20230607_0621      50.0 571.0464 13.3187 4300.3781
## 5 TEMPEST_SW_SOURCE_20230607_1515spk      50.0 707.6693 16.5051 4345.0138
##      Cl_Area  S04_mM      Cl_mM  salinity  S04_ugmL  Cl_ugmL  S04_ugmL_spk
## 1 29.2627 0.3985028 0.6436615 0.04125776 11.614565 20.74349 14.588417
## 2 65.6646 8.0809264 65.6526911 4.20561594 5.181510 46.54777  6.642967
## 3 88.0403 10.4874111 88.0243583 5.63870354 6.724517 62.40927  9.360423
## 4 121.3302 17.8118029 121.3082680 7.77080923 11.420939 86.00754 11.263157
## 5 122.5896 22.0732782 122.5673850 7.85146594 14.153315 86.90029 14.153315
```

```
#now we need to calculate the spike concentration and calculate the spike recovery
```

```
spkconc <- (250)      # in ug
spkvol <- 10          # in uL
spkvol <- spkvol/1000
#spike for these samples was 10uL of the 250mM standard
QAspks$S04_spk_Conc <- (spkconc)*spkvol      # mmoles of S04
head(QAspks)
```

```
##                               Sample_ID Dilution  S04_ppm S04_Area  Cl_ppm
## 1    TEMPEST_FW_SOURCE_20230607_0611      1.1  12.7760  13.5445  22.8178
## 2    TEMPEST_SW_E3_20230607_1600      50.0  259.0745   6.0425  2327.3879
## 3    TEMPEST_SW_I5_20230607_1600      50.0  336.2264   7.8419  3120.4635
## 4    TEMPEST_SW_SOURCE_20230607_0621      50.0  571.0464  13.3187  4300.3781
## 5 TEMPEST_SW_SOURCE_20230607_1515spk      50.0  707.6693  16.5051  4345.0138
##      Cl_Area      S04_mM      Cl_mM  salinity  S04_ugmL  Cl_ugmL  S04_ugmL_spk
## 1  29.2627  0.3985028  0.6436615  0.04125776  11.614565  20.74349   14.588417
## 2  65.6646  8.0809264  65.6526911  4.20561594   5.181510  46.54777    6.642967
## 3  88.0403 10.4874111  88.0243583  5.63870354   6.724517  62.40927    9.360423
## 4 121.3302 17.8118029 121.3082680  7.77080923  11.420939  86.00754   11.263157
## 5 122.5896 22.0732782 122.5673850  7.85146594  14.153315  86.90029   14.153315
##      S04_spk_Conc
## 1          2.5
## 2          2.5
## 3          2.5
## 4          2.5
## 5          2.5
```

```
#need to determine dilution factors and initial amount of sample added
#if your samples are all the same dilution just use the first line
#for Steph / COMPASS this depends on the site so...
QAspks$Dilution <- 1
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 50, QAspks$Dilution)
$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 100, QAspks$Dilution)
$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 50, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 100, QAspks$Dilution)
$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 200, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$Dilution)
QAspks$Dilution <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$Dilution)

#Set Sample volumes in uL
QAspks$SampleVol <- 1
QAspks$SampleVol <- ifelse(QAspks$Dilution == 50, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 1.1, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 5.15, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 4.53, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 56.51, 1483, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 1.25, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 1.1, 1500, QAspks$SampleVol)
QAspks$SampleVol <- ifelse(QAspks$Dilution == 52.74, 1483, QAspks$SampleVol)

# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "TEMPEST"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_UP"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_TR"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_WC"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_PPR_UP"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_PPR_TR"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "MSM_PPR_WC"), 1475, QAspks$SampleVol)
```

```

# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_UP"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_TR"), 1501, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GCrew_WC"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_UP"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_TR"), 1475, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "GWI_WC"), 1462, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_UP"), 100, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_TR"), 100, QAspks$SampleVol)
# QAspks$SampleVol <- ifelse(str_detect(QAspks$Sample_ID, "SWH_WC"), 100, QAspks$SampleVol)

#change sample volume to L
QAspks$SampleVol <- QAspks$SampleVol/1000
head(QAspks)

```

```

##           Sample_ID Dilution  S04_ppm S04_Area  Cl_ppm
## 1  TEMPEST_FW_SOURCE_20230607_0611      1.1  12.7760  13.5445  22.8178
## 2      TEMPEST_SW_E3_20230607_1600     50.0  259.0745   6.0425  2327.3879
## 3      TEMPEST_SW_I5_20230607_1600     50.0  336.2264   7.8419  3120.4635
## 4  TEMPEST_SW_SOURCE_20230607_0621     50.0  571.0464  13.3187  4300.3781
## 5 TEMPEST_SW_SOURCE_20230607_1515spk     50.0  707.6693  16.5051  4345.0138
##      Cl_Area      S04_mM      Cl_mM  salinity  S04_ugmL  Cl_ugmL  S04_ugmL_spk
## 1  29.2627  0.3985028  0.6436615  0.04125776  11.614565  20.74349   14.588417
## 2  65.6646  8.0809264  65.6526911  4.20561594   5.181510  46.54777    6.642967
## 3  88.0403 10.4874111  88.0243583  5.63870354   6.724517  62.40927    9.360423
## 4 121.3302 17.8118029 121.3082680  7.77080923  11.420939  86.00754   11.263157
## 5 122.5896 22.0732782 122.5673850  7.85146594  14.153315  86.90029   14.153315
##      S04_spk_Conc SampleVol
## 1           2.5         1.500
## 2           2.5         1.483
## 3           2.5         1.483
## 4           2.5         1.483
## 5           2.5         1.483

```

```

#gives us the total S04 in the sample in mmoles
QAspks$S04_Total_unspkd <- QAspks$S04_ugmL*QAspks$SampleVol

##total S04 in spiked sample in mmoles
QAspks$S04_Total_spkd <- (QAspks$S04_ugmL_spk)*(QAspks$SampleVol+spkvol)

QAspks$S04_expctd_spkd <- (QAspks$S04_Total_unspkd + QAspks$S04_spk_Conc)
QAspks$spk_recovery <- (QAspks$S04_Total_spkd/QAspks$S04_expctd_spkd)*100
QAspks$S04_spks_flag <- ifelse(QAspks$spk_recovery <=120 & QAspks$spk_recovery >=80 , 'YES', 'NO, rerun')

head(QAspks)

```

```

##           Sample_ID Dilution  S04_ppm S04_Area  Cl_ppm
## 1  TEMPEST_FW_SOURCE_20230607_0611      1.1  12.7760  13.5445  22.8178
## 2      TEMPEST_SW_E3_20230607_1600     50.0  259.0745   6.0425  2327.3879
## 3      TEMPEST_SW_I5_20230607_1600     50.0  336.2264   7.8419  3120.4635
## 4  TEMPEST_SW_SOURCE_20230607_0621     50.0  571.0464  13.3187  4300.3781
## 5 TEMPEST_SW_SOURCE_20230607_1515spk     50.0  707.6693  16.5051  4345.0138
##      Cl_Area      S04_mM      Cl_mM  salinity  S04_ugmL  Cl_ugmL  S04_ugmL_spk

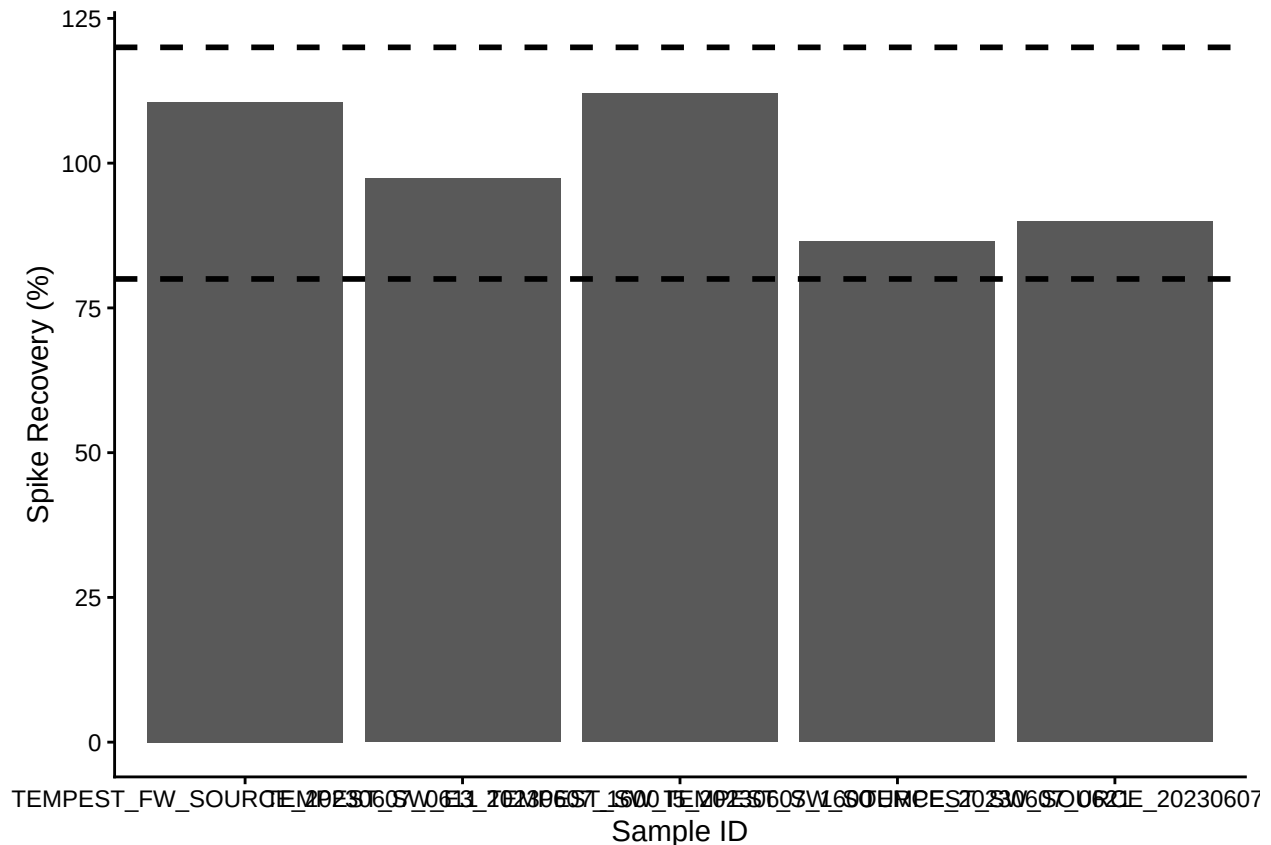
```

```
## 1 29.2627 0.3985028 0.6436615 0.04125776 11.614565 20.74349 14.588417
## 2 65.6646 8.0809264 65.6526911 4.20561594 5.181510 46.54777 6.642967
## 3 88.0403 10.4874111 88.0243583 5.63870354 6.724517 62.40927 9.360423
## 4 121.3302 17.8118029 121.3082680 7.77080923 11.420939 86.00754 11.263157
## 5 122.5896 22.0732782 122.5673850 7.85146594 14.153315 86.90029 14.153315
## S04_spk_Conc SampleVol S04_Total_unspkd S04_Total_spkd S04_expctd_spkd
## 1 2.5 1.500 17.421848 22.02851 19.92185
## 2 2.5 1.483 7.684179 9.91795 10.18418
## 3 2.5 1.483 9.972458 13.97511 12.47246
## 4 2.5 1.483 16.937252 16.81589 19.43725
## 5 2.5 1.483 20.989367 21.13090 23.48937
## spk_recovery S04_spks_flag
## 1 110.57463 YES
## 2 97.38586 YES
## 3 112.04777 YES
## 4 86.51373 YES
## 5 89.95943 YES
```

#plot spk recoveries output as a bar graph to easily check - want any over 10% to be red need to work on

```
spksbar <- ggplot(data = QAspks, aes(x = Sample_ID, y = spk_recovery)) +
  geom_bar(stat = 'identity') +
  scale_fill_manual(values=c("lightblue4")) +
  theme_classic() + labs(x= "Sample ID", y="Spike Recovery (%)") +
  theme(legend.position="none") +
  geom_hline(yintercept=80, linetype="dashed", color = "black", size=1) +
  geom_hline(yintercept=120, linetype="dashed", color = "black", size=1)
```

```
spksbar
```



```
#check for percent of no, reruns to see if it would warrant reruns
Perc_spks <- QAspks %>%
  group_by(S04_spks_flag) %>%
  summarise(no_rows = length(S04_spks_flag))
Perc_spks$Total <- length(QAspks$S04_spks_flag)
Perc_spks$Percent <- (Perc_spks$no_rows / Perc_spks$Total)*100
head(Perc_spks)
```

```
## # A tibble: 1 x 4
##   S04_spks_flag no_rows Total Percent
##   <chr>         <int> <int>   <dbl>
## 1 YES             5     5    100
```

Make final dataframe with IDs

```
#for steph <- pull out identifiers of the sample names
#pull the sample ID and separate it by the underscores
IDs <- data.frame(do.call('rbind', strsplit(as.character(sampledat2$Sample_ID), '_ ', fixed=TRUE)))
colnames(IDs) <- c("Project" , "Plot", "Grid", "Date", "Time")
head(IDs)
```

```
##   Project   Plot   Grid   Date Time
## 1 TEMPEST ESTUARY SOURCE 20230607 0710
```

```
## 2 TEMPEST ESTUARY SOURCE 20230607 1346
## 3 TEMPEST ESTUARY SOURCE 20230607 1700
## 4 TEMPEST      FW      B4 20230607 0800
## 5 TEMPEST      FW      B4 20230607 1600
## 6 TEMPEST      FW      C3 20230607 0800
```

```
#rejoin them to the dataframe
alldat <- cbind(IDs, sampled2)
head(alldat)
```

```
##   Project   Plot   Grid   Date Time   Sample_ID
## 1 TEMPEST ESTUARY SOURCE 20230607 0710 TEMPEST_ESTUARY_SOURCE_20230607_0710
## 2 TEMPEST ESTUARY SOURCE 20230607 1346 TEMPEST_ESTUARY_SOURCE_20230607_1346
## 3 TEMPEST ESTUARY SOURCE 20230607 1700 TEMPEST_ESTUARY_SOURCE_20230607_1700
## 4 TEMPEST      FW      B4 20230607 0800 TEMPEST_FW_B4_20230607_0800
## 5 TEMPEST      FW      B4 20230607 1600 TEMPEST_FW_B4_20230607_1600
## 6 TEMPEST      FW      C3 20230607 0800 TEMPEST_FW_C3_20230607_0800
##   SO4_ppm SO4_Area   Cl_ppm Cl_Area   SO4_mM   Cl_mM   salinity
## 1 581.2849 13.5575 4335.7195 122.3274 18.13115721 122.3052045 7.83467114
## 2 547.2306 12.7632 4145.0144 116.9468 17.06895197 116.9256530 7.49006702
## 3 553.9434 12.9198 4153.6146 117.1895 17.27833437 117.1682539 7.50560758
## 4 11.1494 11.8200 27.9934 35.9001 0.34776669 0.7896587 0.05061007
## 5 2.4624 2.2973 206.4674 233.0096 0.07680599 5.8241862 0.37311259
## 6 10.7750 11.4231 23.9976 30.7756 0.33608858 0.6769422 0.04338966
```

Make final dataframe with IDs

```
write.csv(alldat, file="Processed Data/COMPASS_TEMPEST_Dionex_ClSo4_Final_20230608.csv")
```

#Cha

END